

ANTONIO RAPUANO

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SUMMARY

I'm an engineer driven by curiosity and a strong desire to turn theory into practical systems. I enjoy working across the entire development pipeline – from modeling and simulation to implementation and testing. I'm looking for a role where I can keep learning and contributing to impactful technologies, especially in robotics and autonomous systems.

EXPERIENCE

(Details and media at ntonioa.github.io/projects)

Autopilot Control Systems Engineer

Fall 2024 - Present

Sapienza Flight Team (sasa-aerospace.it/flight-team)

Highlights: Control architecture design • Implementation on custom hardware • Testing and validation

Race Timing Specialist and Team Lead

Fall 2018 - Present

ICRON (icron.it)

Highlights: Race logistics and technical setup • Start/finish line operations • Ranking and award management

EDUCATION

(Details and media at ntonioa.github.io/projects)

M. Sc. in Control Engineering

Fall 2022 - Summer 2025 (expected)

Università degli Studi di Roma "La Sapienza", Rome, Italy

Final grade (expected): 110/110 with honors

Thesis: Modeling, approximation, and nonlinear predictive control of suspended deformable cable continuum dynamics for aerial pick and place (Supervisor: Prof. Antonio Franchi)

Relevant coursework: Analysis, modeling and control of nonlinear, hybrid, multivariable, networked systems • Basic and advanced robotics • Estimation • Optimal control • Machine learning • Robust control

B. Sc. in Electronic Engineering for Automation and Telecommunications

Fall 2018 - Winter 2022

Università degli Studi del Sannio, Benevento, Italy

Final grade: 110/110 with honors

Thesis: Implementation on FPGA of a circuit controlling a mechanical arm (Supervisor: Prof. Marco Pisco)

Relevant coursework: Analog and digital electronics • Analysis, modeling and control of linear systems • Digital design • Electronic measurements • Signal processing • Telecommunications

ACADEMIC PROJECTS

(Details and media at ntonioa.github.io/projects)

- Control of underactuated robots via input-constrained receding-horizon DDP Winter 2024 - Fall 2024
- Fault-tolerant formation control of passive multi-agent systems using energy tanks Summer 2024
- Decentralized PEV charging control based on a subgradient method for MILPs Winter 2023 - Spring 2024
- Modeling and control of the Mars helicopter 'Ingenuity' Fall 2023 - Spring 2024
- Control of a video-game race car using a convolutional neural network Winter 2024
- Self-balancing two-wheeled robot Fall 2023 - Winter 2024
- Signal acquisition through an A/D converter Spring 2021

SKILLS

Software and tools: Arduino • Blender • Fusion 360 • LTSpice • Office • Quartus Prime • Simulink • Stateflow

Programming and formatting: C/C++ • Java • HTML/CSS • LaTeX • MATLAB • Python • Verilog

Hardware: 3D printing • Circuit board soldering

Personal skills: Attention to detail • Autonomy and initiative • Critical thinking • Curiosity and continuous learning • Problem-solving attitude • Project planning • Technical writing • Time management

LANGUAGES

Italian, native

English, fluent

Cambridge English C1 Advanced - Score 207 (Grade A, CEFR Level C2)

Summer 2022